

Tikhon Petrishchev

Embedded Systems Programmer
Moscow, 23 years old

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Key Skills

- Firmware development for real-time embedded systems (C, FreeRTOS, ESP32)
- Reading and designing electrical schematics; development of electronic circuits and PCBs
- Mobile application development; Swift, SwiftUI
- Working with low-level protocols; reverse engineering for device integration; Bluetooth LE
- Web application development in Go and Python; Linux userspace administration and debugging; Docker

Work Experience

TOPODRON, Moscow

February 2026 – August 2026

The company develops and supplies surveying equipment for drones: lidars, SLAM scanners, bathymetric and photogrammetric systems, and PPK GNSS modules.

Embedded Systems Programmer

- Firmware improvements for devices built on single-board computers / Linux, C
- Diagnostics and repair of electronic devices

IPat LLC, Moscow

June 2021 – Present

The company develops Medsenger, a medical system for remote monitoring and doctor–patient communication (chat, intelligent monitoring agents, a medical record with analytics, integration with medical devices, automated questionnaires, monitoring scenarios for various diseases).

Project Author, Architect, Team Lead *August 2024 – October 2025*

- Development of a specialized “Smart Pill Dispenser” integrated with a remote health monitoring and treatment adherence platform, funded by a grant from the Novosibirsk Academgorodok Science and Technology Park Foundation
- Circuit and prototype PCB development / KiCad
  https://github.com/tikhonp/kicad-pill-dispenser
- Server-side development / Go, Echo, Templ, HTMX, SQLX, PostgreSQL
  https://github.com/tikhonp/medsenger-pill-dispenser-bot
- Firmware development / C, ESP-IDF, FreeRTOS, ESP32-C3
  https://github.com/tikhonp/telepat-pill-dispenser
- 3D modeling of enclosures and internal components / SolidWorks

Backend Developer, Flutter Developer *2024*

- Development of an API library for internal microservices in Go: https://github.com/tikhonp/maigo
- Integration of FreeStyle Libre glucose monitors / Go, Echo, Templ, SQLX, PostgreSQL
- Integration of Xiaomi smart scales with the Medsenger remote monitoring service. Flutter-based mobile app for iOS and Android. Integration microservice / Go, Echo, Templ, SQLX, PostgreSQL

Backend Developer *2023*

- Development of a microservice for configurable automated medical phone surveys via the Voximplant service / Python, Django, Django REST Framework, PostgreSQL

Architect, iOS Developer *2023*

- Development of a mobile app integrating with the Axma Hemocard Bluetooth LE tonometer and ECG device / Swift, SwiftUI, Kotlin

Architect, iOS Developer *2022–2024*

- Development of the Medsenger iOS mobile app (~60k lines) / Swift, SwiftUI, CoreData, HealthKit, WebRTC video calls

Reverse Engineer, iOS Developer *2022*

- Integration of the “Contec” Bluetooth spirometer into the medical messenger; reverse engineering of the Bluetooth LE protocol
- Development of an iOS mobile application for the Contec spirometer / Swift, SwiftUI, CoreBluetooth
- Backend for the integration microservice using Flask

iOS Developer *2021*

- Integration of the “Serdechko” cardiograph into the Medsenger medical messenger
- Patching the mobile app on iOS and Android
- Backend for the microservice using Flask

Developer *2019–2021*

- Embedded software for a smart speaker integrated with a medical service; development of a library for Yandex SpeechKit / Python, Yandex SpeechKit / https://github.com/tikhonp/yandex-speechkit-lib-python
- Backend / Django, Django REST Framework, PostgreSQL
- iOS application / Swift, SwiftUI

Education

MIPT, Zhukovsky

Master’s degree, started 2026
Software Engineering, Faculty of Aeromechanics and Flight Engineering

RTU MIREA, Moscow

Bachelor’s degree, Class of 2026
Industrial Informatics

Achievements

Authorship of Computer Programs

- Certificate of state registration of a computer program No. 2024611961 “Medsenger.Prenatal – subsystem for remote monitoring and informational support for pregnant women” dated January 26, 2024
- Certificate of state registration of a computer program No. 2024611962 “Medsenger.Hyper – subsystem for remote blood pressure monitoring using various measurement data transfer methods” dated January 26, 2024
- Certificate of state registration of a computer program No. 2024616051 “Oncofertility Registry” dated March 15, 2024

Competitions

- Winner of the YoHack hackathon by CROC, April 2020

Interests and Hobbies

- Classical and jazz music, piano, painting
- Sport orienteering, running